

SeokHyeon (Lucas) Kong

CONTACT INFORMATION	Moore 1st floor University of Pennsylvania 200 South 33rd Street, Philadelphia, PA 19104	Personal Website: lucasshkong.github.io Email: lucasshkong@gmail.com
RESEARCH INTERESTS	My research interests lie in <i>Quantum Computer Architecture</i> , currently focusing on Quantum Error Correction.	
EDUCATION	University of Pennsylvania , Philadelphia, PA • Ph.D. in Computer and Information Science • Advisor: Prof. Gushu Li	Aug 2026
	Sungkyunkwan University (SKKU) , Suwon, Korea • B.S. in Electronic and Electrical Engineering • B.S. in Advanced Semiconductor Engineering – System Architecture Track • <i>Graduated with Summa Cum Laude</i> (GPA: 4.36 / 4.5)	Mar 2020 – Feb 2026
	The Pennsylvania State University , University Park, PA • Exchange Student in Computer Engineering • <i>Term Honor: Dean's List</i> (Grade: 4.0 / 4.0)	Fall 2024
PUBLICATIONS	[1] Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy SeokHyeon Kong, Donghwan Kim, Euiseong Seo, and Kiwan Maeng CAL 2025 – IEEE Computer Architecture Letters	
	[2] Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system SeokHyeon Kong and Hayoung Oh JKIICE 2023 – Journal of the Korean Institute of Information and Communication Engineering	
POSTER PRESENTATION	[3] Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices SeokHyeon Kong, Chanhee Park, Jang-il Moon, and Denny Dahl Qcenter – Quantum Information Research Support Center, Korea	
EXPERIENCE	Hanwha Systems , Seongnam, Korea <i>Software Engineer in the Avionics System R&D Team</i> • Collaborated with The Boeing Company on ELAD (Eagle Large Area Display) for F-15 aircraft.	Jan 2026 – Jul 2026
	IonQ , Remote <i>Research Mentee with Dr. Denny Dahl</i> • Addressed the Instant Insanity problem by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods [3]. • Solved the Four Corners Map Coloring problem using a Quantum Approximate Optimization Algorithm (QAOA) with two different encoding schemes [3].	Aug 2025 – Sep 2025

The Pennsylvania State University, University Park, PA Aug 2024 – Sep 2025
Undergraduate Researcher with Prof. Kiwan Maeng

- Conducted GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differential privacy [1].
- Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality [1].

TECHNICAL PROJECT

SoC Design and Practice, SKKU
Graduate Level Coursework

Fall 2025

- Designed and implemented a multi-channel, pipelined DMAC engine and cache controller in SystemVerilog with AMBA APB-based configuration and AXI-based data transfer.
- Synthesized and verified the design using Synopsys Design Compiler, optimizing area and timing to meet target performance specifications.

TEACHING EXPERIENCE

Academic Tutor, SKKU
Intro to Electromagnetism

Fall 2023

- Led weekly Q&A sessions and problem-solving discussions.

Teaching Assistant, SKKU
Engineering Mathematics 2

Summer 2023

- Prepared and organized class materials, including exams.

HONORS AND AWARDS

Graduate Fellowship and Research Stipend Award, UPenn

2026

Summa Cum Laude, SKKU

2026

Director's Award – Quantum Hackathon, QCenter

2025

President's List, SKKU

2024, 2025

Dean's List, SKKU

SP2023, FA2023, SP2024, FA2024, SP2025

Korea-U.S. Student Exchange Scholarship

FA2024

in the field of high-tech industry (\$9,000), KIAT

Student Success Scholarship, SKKU

FA2023

SKILLS

Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX

Hardware: System Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute

LEADERSHIP AND SERVICE

Executive Board, President's List (Honor Society at SKKU)

2025

Qiskit Advocate, IBM Quantum

2025

Squad Leader, Republic of Korea Army (Military Service) Apr 2021 – Oct 2022

REFERENCES

Gushu Li, Ph.D.

Email: gushuli@seas.upenn.edu

Assistant Professor of Computer and Information Science
University of Pennsylvania

Kiwan Maeng, Ph.D.

Email: kvm6242@psu.edu

Assistant Professor of Computer Science and Engineering
The Pennsylvania State University

Denny Dahl, Ph.D.

Email: denny.dahl@ionq.co

Customer Success Director
IonQ